Change Detection in SAR Images Using Hypothesis Testing and Shannon Entropy Based on the Rayleigh Distribution

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***Abstract*—This article presents a change detection algorithm based on Shannon entropy and the Rayleigh distribution for both single-look and multilook synthetic aperture radar (SAR) images. While hypothesis testing has been widely used in SAR change detection, few studies have applied this approach to both types of data. To address this, we propose a method that utilizes Shannon entropy to detect changes between two samples. The performance of the algorithm was evaluated through Monte Carlo simulations using synthetic SAR data and further validated on real-world datasets, including single-look images from the CARABAS II dataset and multilook data from the uninhabited aerial vehicle synthetic aperture radar (UAVSAR). The results demonstrate that the proposed method is effective in detecting changes. This article highlights the versatility of the approach, which is capable of handling both single-look and multilook SAR data, and reinforces the way for future research into alternative entropy measures and probability distributions in change detection tasks.**

***Index Terms*—Change detection, hypothesis testing, Rayleigh distribution, Shannon entropy, synthetic aperture radar (SAR) images.**

I. Introduction

REMOTE sensing techniques are tools for the acquisition of data about the Earth's surface without the need for direct physical contact. Sensors employed in these methods are commonly installed on platforms such as satellites, aircraft, drones, and ships, enabling efficient large-scale data collection [1]. A prominent system within remote sensing is synthetic aperture radar (SAR), an active instrument that generates its own illumination [2]. SAR creates images by analyzing the interaction between the emitted energy and the energy reflected back from the target. The resulting images are influenced by the target's physical characteristics, such as shape and texture [3].

An interesting advantage of SAR is its ability to penetrate clouds, rain, and vegetation, allowing it to operate effectively in various weather conditions. This capability has led to SAR sensors being characterized as all-weather instruments proficient at generating high-resolution images. However, SAR systems can also produce lower resolution images, which may be sufficient or preferable for specific applications, such as wide-area surveillance, rapid change detection, or when data transmission and storage resources are limited. This flexibility makes SAR a versatile tool across various spatial and temporal scales. The high resolution is achieved through broad bandwidth, which refers to the frequency and wavelength at which the radar operates [3].

The microwaves transmitted and received by SAR sensors are emitted as pulses and can exhibit either circular [4] or linear polarizations [5]. These polarizations can be aligned in parallel, such as horizontal–horizontal (HH) and vertical–vertical (VV), or cross-polarized (HV and VH), where the first letter represents the transmitted polarization and the second denotes the received polarization. When only a single pair of polarization directions is used, the resulting images are classified as single-polarization SAR [6].

Despite the aforementioned advantages, SAR systems rely on coherent illumination for image formation. This coherent nature, while essential for image formation processing, also leads to the phenomenon of speckle noise. Speckle is a form of multiplicative noise inherent in coherent imaging systems, arising from the random constructive and destructive interference of scattered signals from multiple unresolved scatterers within a single resolution cell [7]. This interference is a consequence of small variations in surface height within the resolution cell, relative to the radar wavelength, causing random phase differences in the backscattered waves. The subsequent image formation process involves steps such as demodulating the complex return signal (which already contains speckle) into in-phase (I) and quadrature (Q) components, sampling, and digital conversion to create a data sequence [8]. The presence of speckle in the received signal results in a characteristic granular appearance in the final SAR images, complicating their interpretation and analysis.

To mitigate speckle effects, a common approach in SAR processing is the multilook technique. Unlike single-look processing, where a scene is captured and analyzed in a single synthetic aperture, multilooking divides the radar return signal into multiple subapertures (or "looks") during processing [9]. These subapertures are generated from distinct portions

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of the Doppler or azimuth spectrum of the same acquired dataset [10], effectively averaging independent measurements of the scene. This averaging reduces speckle by improving the signal-to-noise ratio [2]. However, the multiplicative nature of speckle in SAR data often causes deviations from the behavior typically associated with a Gaussian distribution. Indeed, the Gaussianity assumption has been extensively adopted in statistical signal processing [11], remote sensing applications [12], [13], [14], and SAR image modeling and detection [15], [16], [17], [18]. However, real-world measured signals often exhibit non-Gaussian characteristics [19], [20], including asymmetry and strictly positive values, such as amplitude data of SAR image pixel values [2]. When aiming to model this SAR data, a distribution widely applied in signal and image processing is the Rayleigh distribution [21], [22], [23]. The Rayleigh distribution is particularly relevant for SAR image modeling, as it effectively characterizes image pixel amplitude values [24], [25], [26], [27], in particular, this distribution is known to well fit SAR data homogeneous areas [2], [28]. The main justification for supposing the Rayleigh distribution as a model for the amplitude of SAR data is that the assumption of a large number of reflectors in an observed image allows one to invoke the central limit theorem, according to which the distribution of the real and complex parts of the received signal are independent and approaches the Gaussian distribution with zero mean and constant variance [25], [28]. Thus, under these assumptions, the amplitude values of complex SAR data are exactly Rayleigh distributed.

Numerous methods for processing SAR images have been proposed across various fields, including: 1) classification [29]; 2) segmentation [30]; 3) edge detection [31]; and 4) change detection [32], [33]. Change detection aims explicitly to identify differences between two or more images of the same scene captured at different times. By detecting changes in the characteristics of the Earth's surface, these methods provide valuable insights into both human and natural phenomena, facilitating improved management and utilization of resources [34].

Significant contributions have been made by researchers to the field of change detection. For instance, [35] utilized techniques to detect temporal changes in SAR images based on differences in signal magnitudes between the two images. In addition, [36] introduced a change detection algorithm for SAR images employing the $\mathcal{G}^0$ distribution and Kullback–Leibler divergence.

Information theory has been shown to be an important tool for statistical inference [37] and image processing [38]. The concepts of information and entropy were formally introduced in the context of communication theory by Pardo [39] and have become foundational across various fields. Entropy, in particular, has emerged as a core concept with diverse interpretations, represented as follows: 1) disorder in statistical mechanics [37] and 2) diversity in economics and biology [39] and, often measured through probability distributions.

Research on SAR data [36], [40], [41], [42], [43] has shown promising results in exploring the relationship between information theory measures and SAR data models. Hypothesis testing has been increasingly utilized in SAR image change detection, as discussed in works like [44], [45], [46]. In the case of polarimetric SAR images, studies have employed hypothesis tests and information theory for change detection (e.g., [32]). However, to the best of our knowledge, no studies have yet applied these approaches to amplitude data and single-look SAR images. In this context, we propose a change detection method based on Shannon entropy and the Rayleigh distribution. While the Rayleigh model is theoretically appropriate for physically single-look SAR images due to its fit to homogeneous regions, prior works [47], [48] have shown that it also performs well in multilook data. Motivated by this, our aim is not to derive a method tied strictly to the distributional assumptions of either single-look or multilook data but rather to demonstrate a robust approach capable of delivering effective results across both types of SAR imagery.

This article aims to fill that gap by presenting a novel topic treatment. Specifically, we propose a change detection algorithm based on Shannon entropy for the Rayleigh distribution. Monte Carlo simulations were conducted to assess the performance of the proposed method on synthetic SAR data, and two experiments with real-world data were carried out to validate the approach. The first real-data experiment utilizes single-look images from the CARABAS II dataset, widely studied for detecting military vehicles concealed by forests [45], [49], [50], [51], [52]. In addition, to demonstrate the robustness of the proposed model, we incorporate a multilook image obtained from the uninhabited aerial vehicle synthetic aperture radar (UAVSAR) [53], previously used in a change detection method proposed for single-look polarimetric data based on geodesic distance techniques.

This article is organized as follows. In Section II, the Rayleigh distribution, your maximum likelihood estimation process, and the convergence to the normal distribution are described. Section III presents the results of Shannon entropy for the Rayleigh distribution and the proposed hypothesis test in this case. The change detection process using the proposed hypothesis test is found in Section IV. Numerical results with analyses with simulations and real SAR data are presented in Section V. Finally, Section VI provides the study's conclusion.

II. Statistical Modeling for Single-Look SAR Data—The Rayleigh Distribution

The Rayleigh distribution is a continuous probability distribution applicable to positive-valued random variables. It finds extensive application across various domains, including signal processing, radar technology, and communication systems [2], [28]. In the context of SAR images, backscatter data conveys crucial information about the mapped area, reflecting the physical characteristics of the target, such as moisture content and topography [2], [28].

A. Model

Let Y be a Rayleigh-distributed random variable with a scale parameter $\beta > 0$. The probability density function (pdf) of the Rayleigh distribution is given by [26]

$$f_Y(y;\beta) = \frac{y}{\beta^2} e^{-\frac{y^2}{2\beta^2}}, \quad y \geq 0. \tag{1}$$

The expectation and the variance of Y are given by

$$\mathrm{E}(Y) = \beta\sqrt{\frac{\pi}{2}}$$
$$\mathrm{Var}(Y) = \beta^2\left(\frac{4-\pi}{2}\right).$$

1) Parameter Estimation Process: Let $y_1, y_2, \ldots, y_N$ be N independent random samples, where each sample y_n follows the Rayleigh density presented in (1). The estimated parameter of the Rayleigh distribution can be performed by the maximum likelihood method [54]. The likelihood function for β based on y_n is defined as

$$L(\beta; y_n) = \prod_{n=1}^{N} \frac{y_n}{\beta^2} e^{-\frac{y_n^2}{2\beta^2}}.$$

The maximum likelihood estimator (MLE) is given by $\widehat{\beta}$, which is the argument that maximizes

$$\widehat{\beta} = \arg\max_{\beta} \ell(\beta)$$

being $\ell(\beta)$ the log-likelihood function of the parameter for the observed signal. The quantity $\ell(\beta)$ is the logarithm of $L(\beta; y_n)$ and is given by $\ell(\beta) = \sum_{n=1}^{N} \ell_n(\beta)$, being

$$\ell_n(\beta) = \log(y_n) - 2\log(\beta) - \frac{y_n^2}{2\beta^2}.$$

The score function denoted by U_β is obtained by differentiating the log-likelihood with respect to β, and it is defined as

$$\frac{d\ell_n(\beta)}{d\beta} = \frac{y_n^2}{2\beta^3} - \frac{2}{\beta}.$$

The MLE for the Rayleigh distribution parameter is obtained by solving the following nonlinear system $U_\beta = 0$, and it is given by:

$$\hat{\beta} = \sqrt{\frac{1}{2N}\sum_{n=1}^{N} y_n^2}. \tag{2}$$

2) Large Data Record Results: Under certain mild regularity conditions [55], an MLE is both consistent and asymptotically normally distributed as the sample size N approaches infinity. Consequently, for large datasets, we have

$$\widehat{\beta} \xrightarrow[N\to\infty]{\mathcal{D}} \mathcal{N}\left(\beta, \mathcal{K}(\beta)^{-1}\right)$$

where $\mathcal{N}(\beta, \mathcal{K}(\beta)^{-1})$ is the Gaussian distribution with mean β and variance $\mathcal{K}(\beta)^{-1}$, being $\mathcal{K}(\beta)^{-1}$ the inverse of the Fisher information matrix, and "$\xrightarrow{\mathcal{D}}$" denoting *convergence in distribution*. This asymptotic distribution can be utilized to construct confidence intervals [54] and to perform hypothesis tests [54].

To derive the Fisher information matrix, we must compute the expectation of the negative second-order partial derivatives of the log-likelihood function [54]. The second-order derivative of $\ell(\beta)$ with respect to β is expressed as follows:

$$\frac{d^2\ell(\beta)}{d\beta d\beta} = \sum_{n=1}^{N} \frac{d}{d\beta}\left(\frac{d\ell_n(\beta)}{d\beta}\right) = \sum_{n=1}^{N} \frac{2}{\beta^2} - \frac{3y_n^2}{\beta^4}. \tag{3}$$

Taking the negative expectation of (3), the Fisher information matrix for the Rayleigh distribution is given by $\mathcal{K}(\beta) = N(4/\beta^2)$, since $\mathrm{E}(Y^2) = 2\beta^2$.

III. Shannon Entropy Based for the Rayleigh Distribution

An SAR image can be interpreted as a collection of regions, in that each potentially can be governed by a different probability distribution [56]. Some statistical approaches have addressed the challenge of accurately distinguishing these regions within an image. One practical approach is hypothesis testing, which enables the assessment of differences in mean amplitude between two distinct regions in an image [56], [57]. In SAR image analysis, this method is useful for identifying land cover types, detecting changes in land cover, and performing classification tasks, as demonstrated in [58] and [59].

Entropy-based measures contribute to image processing tasks, including classification, segmentation, change detection, and others [9]. Information theory in SAR images stands out for its ability to characterize scattering patterns and disorder through metrics such as Shannon Entropy. By quantifying uncertainty, entropy enables the identification of homogeneous and heterogeneous regions within an image, highlighting the distinctive features of each area.

This section summarizes the most important concepts of Shannon entropy based on the Rayleigh distributions. In addition, large data record results for the entropies are revised.

A. Class of h-ϕ Entropies

Pardo [39] introduced a class generalizes the original concept of entropy and proposes to study its asymptotic distribution, denomined the (h, ϕ)-entropy. The utilization of the (h, ϕ)-entropy class will be important for demonstrating the distributional convergence of the entropy, which the result will consequently be used in the construction of the new proposal in this article.

Let Z be a random variable with a pdf $f_Z(z; \boldsymbol{\theta})$, where $\boldsymbol{\theta} \in \boldsymbol{\Theta} \subseteq \mathbb{R}^p$ represents the parameter vector of the distribution, with p denoting the dimension of the parameter space. The class (h, ϕ)-entropy of Z is defined by

$$H_\phi^h(\boldsymbol{\theta}) = h\left(\int_{\mathcal{A}} \phi(f_Z(z; \boldsymbol{\theta}))\, dz\right)$$

where either $\phi : [0, \infty) \to \mathbb{R}$ is concave and $h : \mathbb{R} \to \mathbb{R}$ is increasing, or ϕ is convex and h is decreasing. In particular, the Shannon entropy is obtained when $h(z) = z$ and $\phi(z) = -z\log(z)$.

The following result establishes a foundation for exploring asymptotic statistical inference methods based on entropy. Although our proposal is based on Shannon entropy, in *Lemma 1*, we present the asymptotic convergence in distribution results for the entire $(h - \phi)$ class. These results are fundamental and will be employed in the development of our proposed change detector, which is constructed using a hypothesis test with Shannon entropy based on the Rayleigh distribution.

Lemma 1: Let $\widehat{\boldsymbol{\theta}} = [\widehat{\theta}_1, \widehat{\theta}_2, \ldots, \widehat{\theta}_p]^\top$ be the maximum likelihood estimator for $\boldsymbol{\theta} = [\theta_1, \theta_2, \ldots, \theta_p]^\top$ based on a N-points random sample, $Z_1, \ldots, Z_N$, under the pdf $f(z; \boldsymbol{\theta})$. Then,

$$\sqrt{N}\left[H_h^\phi(\widehat{\boldsymbol{\theta}}) - H_h^\phi(\boldsymbol{\theta})\right] \xrightarrow[N\to\infty]{\mathcal{D}} \mathcal{N}\left(0, \sigma_\phi^2(\boldsymbol{\theta})\right). \tag{4}$$

In addition,

$$\sigma_{\phi}^{2}(\boldsymbol{\theta}) = \boldsymbol{\delta}^{\top} \mathcal{K}(\boldsymbol{\theta})^{-1} \boldsymbol{\delta} \tag{5}$$

where $\mathcal{K}(\boldsymbol{\theta}) = E\{-\partial^2 \log f_Z(Z;\boldsymbol{\theta})/\partial\boldsymbol{\theta}\,\partial\boldsymbol{\theta}^{\top}\}$ is the Fisher information matrix, and $\boldsymbol{\delta} = [\delta_1, \delta_2, \ldots, \delta_p]^{\top}$ such that $\delta_i = \partial H_h^{\phi}(\boldsymbol{\theta})/\partial\theta_i$ for $i = 1, 2, \ldots, p$ [39].

In what follows, we derive some properties for the Rayleigh distribution, which are needed for hypothesis tests.

B. Entropy-Based Inference for the Rayleigh Distribution

The differential entropy, introduced by Shannon [60], quantifies the uncertainty inherent in a continuous probability distribution. Let $Y \sim \mathcal{R}(\beta)$, where $\mathcal{R}(\beta)$ denotes a Rayleigh distribution with parameter β, its Shannon entropy for Y is given by Nielsen [61]

$$H_S(\mathcal{R}) = H_S\left([\beta]\right) = 1 + \log\left(\frac{\beta}{\sqrt{2}}\right) + \frac{\gamma}{2} \tag{6}$$

where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant. In this article, we assume the term "stochastic entropy" refers to the expression of entropy after replacing the model parameters with their MLEs.

To study the asymptotic distribution of the Rayleigh stochastic Shannon entropy, say $\widehat{H}_S(\mathcal{R}) = H_S([\hat{\beta}])$, which is the result of (6) using the estimative $\hat{\beta}$ obtained with (2). The following result gives theoretical quantities to define the variance of $\widehat{H}_S(\mathcal{R})$.

Proposition 1: Let $Y \sim \mathcal{R}(\beta)$. Define

$$\boldsymbol{\delta}_{\mathcal{R}} = \delta_{\beta}^{\mathcal{R}} := \frac{\partial H(\mathcal{R})}{\partial\beta} \text{ in (5).} \tag{7}$$

The expression for $\delta_{\beta}^{\mathcal{R}}$ is

$$\delta_{\beta}^{\mathcal{R}} = \frac{\partial}{\partial\beta}\left[1 + \ln\left(\frac{\beta}{\sqrt{2}}\right) + \frac{\gamma}{2}\right] = \frac{1}{\beta}.$$

To determine the variances of $\widehat{H}(\mathcal{R})$ from (5), we need to find the element of the Fisher information for the Rayleigh distribution, which was defined in Section II.

According to [32], the methodology for hypothesis testing and constructing confidence intervals based on Shannon's entropy is as follows. Let $\boldsymbol{\theta}_1 \in \boldsymbol{\Theta}_1 \subseteq \mathbb{R}^p$ and $\boldsymbol{\theta}_2 \in \boldsymbol{\Theta}_2 \subseteq \mathbb{R}^p$ be two parameter vectors, so

$$\begin{cases} \mathcal{H}_0: \; H_S(\boldsymbol{\theta}_1) = H_S(\boldsymbol{\theta}_2) \\ \mathcal{H}_1: \; H_S(\boldsymbol{\theta}_1) \neq H_S(\boldsymbol{\theta}_2) \end{cases} \tag{8}$$

which represents the following question: "Is there any statistical evidence for rejecting the assumption that two SAR samples come from the same model?" Let $\widehat{\boldsymbol{\theta}}_i$ be the MLE for $\boldsymbol{\theta}_i$ based on a N-points random sample, Z_i for $i = 1, 2$. So, from Lemma 1, it follows that:

$$\sum_{i=1}^{2} \frac{N_i (H_S(\widehat{\boldsymbol{\theta}}_i) - \bar{\nu})^2}{\sigma_S^2(\widehat{\boldsymbol{\theta}}_i)} \xrightarrow[N_i \to \infty]{\mathcal{D}} \chi_1^2 \tag{9}$$

where χ_1^2 represents the chi-squared distribution with one degree of freedom and

$$\bar{\nu} = \left[\sum_{i=1}^{2} \frac{N_i}{\sigma_S^2\left(\widehat{\boldsymbol{\theta}}_i\right)}\right]^{-1} \sum_{i=1}^{2} \frac{N_i H_S\left(\widehat{\boldsymbol{\theta}}_i\right)}{\sigma_S^2\left(\widehat{\boldsymbol{\theta}}_i\right)}.$$

Then, the following test statistic is obtained:

$$S_S\left(\widehat{\boldsymbol{\theta}}_1, \widehat{\boldsymbol{\theta}}_2\right) = \sum_{i=1}^{2} \frac{N_i \left(H_S\left(\widehat{\boldsymbol{\theta}}_i\right) - \bar{\nu}\right)^2}{\sigma_S^2\left(\widehat{\boldsymbol{\theta}}_i\right)}. \tag{10}$$

This asymptotic convergence result in (9) is supported by the general theory of divergence-based test statistics, as detailed in Pardo's work [39, Th. 3.2]. Under standard regularity conditions, and assuming the consistency and asymptotic normality of the estimators involved, such statistics converge in distribution to a chi-squared distribution with degrees of freedom determined by the number of constraints—in this case, one. The result justifies the use of the statistic in (10) for hypothesis testing purposes.

Proposition 2: Let N_i, $i = 1, 2$, be sufficiently large. If $S_S(\widehat{\boldsymbol{\theta}}_1, \widehat{\boldsymbol{\theta}}_2) = s$, then the null hypotesis $\mathcal{H}_0$ can be rejected at a level α if $P(\chi_1^2 > s) \leq \alpha$. When comparing two samples of the same size ($N_1 = N_2 = N$), (10) reduces to

$$S_S\left(\widehat{\boldsymbol{\theta}}_1, \widehat{\boldsymbol{\theta}}_2\right) = N \frac{\left[H_S\left(\widehat{\boldsymbol{\theta}}_1\right) - H_S\left(\widehat{\boldsymbol{\theta}}_2\right)\right]^2}{\sigma_S^2\left(\widehat{\boldsymbol{\theta}}_1\right) + \sigma_S^2\left(\widehat{\boldsymbol{\theta}}_2\right)} \tag{11}$$

with $\widehat{\boldsymbol{\theta}}_1 = \hat{\beta}_1$ and $\widehat{\boldsymbol{\theta}}_2 = \hat{\beta}_2$, and $\sigma_S^2(\hat{\beta}_i) = (1/4)$, for $i = 1, 2$, in this case.

IV. Change Detection Process

This section presents the algorithm scheme for a new change detector. The proposed change detection method requires at least two images of the same region taken at two different periods with the same dimensions and physical capture characteristics. The detection process examines the difference at each pixel in the image by sliding over all pixels, using a window around each analyzed pixel and its nearest neighbors, where the window size is a hyperparameter.

A pseudocode of the proposed method is shown in Algorithm 1. In summary, samples are taken from the images using a 3×3 window around the pixel under analysis [32], [62], and parameters are estimated to apply the hypothesis test described in Section III. Generally, hypothesis tests in this context are conducted using a nominal level (denoted by α), which represents the Type I error rate, that is, the probability of a false positive (incorrectly identifying a change where none exists). Nascimento et al. [32] conducted a study on p-values in SAR/PolSAR images and, based on their empirical findings, adopted a nominal level of $\alpha = 1 \times 10^{-4}$ to ensure a strict control of false alarms. In the present study, we use a more permissive value of $\alpha = 5 \times 10^{-4}$ (i.e., 0.05%), which provides a balanced tradeoff between sensitivity (ability to detect changes) and specificity (avoiding false positives). This

Algorithm 1 Proposed Change Detection Method Step-by-Step Based on the Shannon Entropy and the Rayleigh Distribution

start:
  for j-th observation:
    Get samples $\mathbf{y}_i^{(j)} = [y_1, \ldots, y_{N_j}]^\top$, where $i = 1, 2$ represents the i-th image, j is the position of the j-th pixel, and $y_1, \ldots, y_{N_j}$ are the samples including the j-th pixel and its neighbors;
    Estimate $\beta_i^{(j)}$ based on $\mathbf{y}_i^{(j)}$, for $i = 1, 2$;
    Calculate $S_S^{(j)}(\widehat{\boldsymbol{\theta}}_1, \widehat{\boldsymbol{\theta}}_2)$, for $\widehat{\boldsymbol{\theta}}_1 = \hat{\beta}_1^{(j)}$ and $\widehat{\boldsymbol{\theta}}_2 = \hat{\beta}_2^{(j)}$, eq. (11);
    Make a decision based on the nominal level α: if $P(\chi_1^2 > s^{(j)}) \leq \alpha$, then the pixel is marked as changed, where $s^{(j)} = S_S^{(j)}(\widehat{\boldsymbol{\theta}}_1, \widehat{\boldsymbol{\theta}}_2)$;
  end
  repeat the process for all observations of the image, i.e., $j = 1, \ldots, N$, where N is the dimension of the image, rows by columns;
end

value is still conservative and can be adjusted depending on the application's tolerance for false alarms. Within our proposal, the significance level α is a tunable parameter that does not alter the core algorithm (see Algorithm 1) but affects the final decision criterion: pixels with test statistics yielding p-values below α are marked as changed. These are shown in white in the output image, while pixels with p-values above α are shown in black. The illustration in Fig. 1 shows the estimation process using a 3×3 window, including a central pixel and its nearest neighbors, totaling nine observations (pixels) in this case. This signal is used to estimate the parameters $\hat{\beta}_i^{(j)}$, for $i = 1, 2$ and $j = 1, \ldots, N$, where N is the total number of pixels in the image. These estimated parameters are then used to calculate the $s^{(j)}$ statistic.

The change detection process is carried out for each pixel in the image using a sliding window approach, moving pixel-by-pixel to check for differences (or changes) in each pixel between the two (or more) analyzed images.

V. Numerical Experiments

In this section, the results of the performance evaluation of the proposed methodology are presented of the proposed methodology through synthetic SAR data (under the Rayleigh distribution) and two experiments with real data. First, Monte Carlo experiments are conducted to measure the nominal level (probability of false alarm-PFA) and probability of detection-PD. The measured SAR data experiment evaluates the method proposal in change detection.

A. Synthetic Data Modeling

The simulations were based on a cropped portion of an image captured by the airborne CARABAS II system, a Swedish ultrawideband (UWB) very-high frequency (VHF) SAR device that operate at horizontal (HH) polarization.

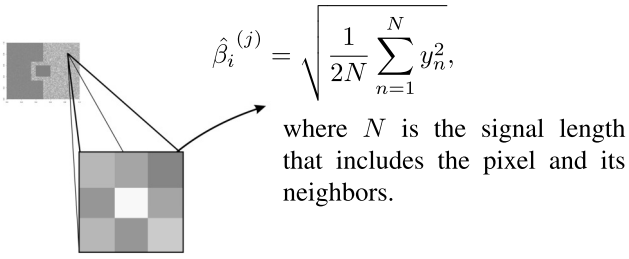
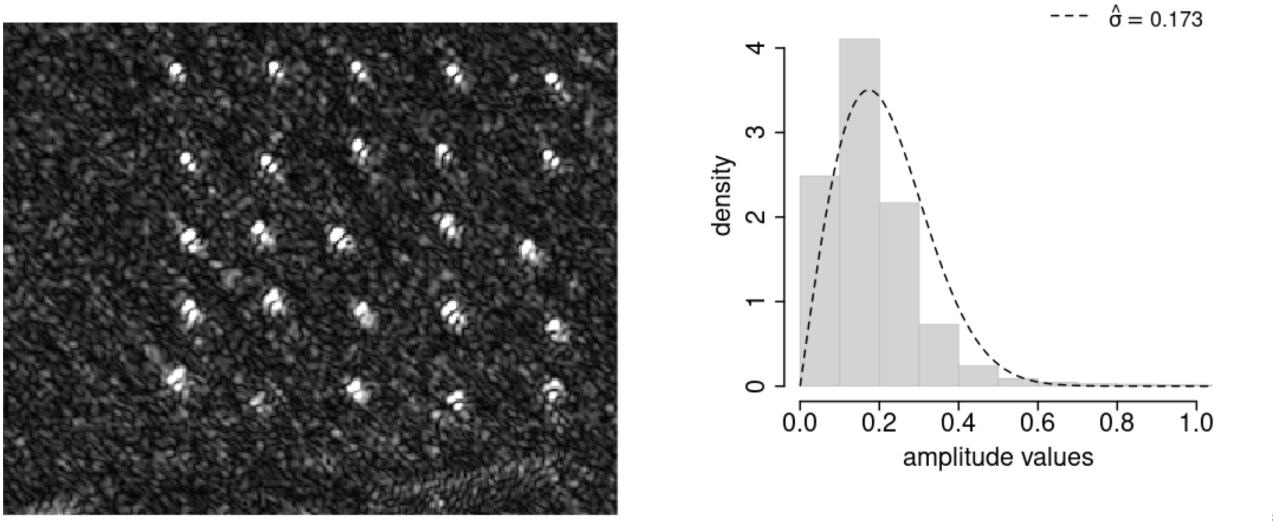


Fig. 1. Visualization of a 3×3 window for the $\hat{\beta}_i^{(j)}$ estimation, using the sample at the jth position and its neighboring pixels for the ith image.

Details related to the image can be found in [50] and [51]. The cropped ground scene image is composed of pine forests and military vehicles.

The cropped image was used to estimate the parameter for use in simulations. The idea was to have the parameter estimation based on real-world data, ensuring that the estimated parameter from this scene could be employed as an input value in the simulations. The area used can be visualized in Fig. 2(a) and has dimensions of 241×301 pixels, and more details about this image can be found in the following section. Fig. 2(b) presents the histogram with the Rayleigh density curve based on the estimated parameter ($\hat{\sigma} = 0.173$). The Rayleigh distribution curve effectively captures the behavior of the data, even when considering an area with diverse characteristics, as mentioned in this cropped area, which contains pine forests (gray and black pixels) and military vehicles (white pixels).

To verify the rejection rate of the proposed hypothesis test as a change detector, a Monte Carlo simulation study with 5000 replications was conducted. Signals were generated with the sample lengths of 9, 25, 49, and 81, which represent the number of observations when considering window sizes around the pixel of 3×3, 5×5, 7×7, 9×9, and 11×11, respectively. For each signal (sample) generated, the test was applied to verify the rejection of the null hypothesis, i.e., to evaluate the Type I error. For each of the 5000 replications, we applied the test under the null hypothesis and recorded whether $\mathcal{H}_0$ was rejected at nominal levels of 10%, 5%, and 1%. The percentages reported in Table I represent the empirical Type I error rates. These values closely match the nominal levels, validating the test's reliability. For instance, a value of 0.0534 for a 5% level indicates a 5.34% rejection rate across 5000 simulations, indicating good alignment with the nominal significance level. These block lengths are commonly used as window sizes in SAR image processing [48], [63], [64], [65]. The values reported in Table I represent the empirical Type I error rates—i.e., the proportion of rejections at each level across all replications. The results in Table I correspond to simulations of Type I error for different sample sizes and nominal levels, including the arithmetic mean of the S_s value ($\overline{S_S}$). These values of $\overline{S_S}$ are expected to converge to 1 as the number of replications increases, given that under the null hypothesis, the distribution of the S statistic converges to χ_1^2. The observed small deviations from 1 are within the range



(a) (b)

Fig. 2. (a) Selected region from a single-look CARABAS II SAR image, including forest and military vehicle targets, used for parameter estimation. (b) Histogram of pixel amplitudes from the selected region overlaid with the fit Rayleigh distribution, demonstrating a good visual fit even in a heterogeneous area.

TABLE I

TYPE I ERROR RATES AND MEAN S_S STATISTICS ($\overline{S_S}$), HERE $\overline{S}_S = \left(\sum_{i=1}^{NC} S_s[i]/NC\right)$, WITH $NC = 5000$ BEING THE NUMBER OF MONTE CARLO REPLICATIONS UNDER THE NULL HYPOTHESIS. THE OBSERVED REJECTION RATES ALIGN CLOSELY WITH THE NOMINAL LEVELS, VALIDATING THE ACCURACY OF THE PROPOSED TEST

Signal length	10% Level (0.10)	5% Level (0.05)	1% Level (0.01)	$\overline{S_S}$ value
9	0.1034	0.0501	0.0111	1.0295
25	0.1036	0.0514	0.0124	1.0341
49	0.1064	0.0531	0.0131	1.0317
81	0.0948	0.0472	0.0096	0.9885
121	0.0971	0.0534	0.0118	0.9885

of acceptable Monte Carlo variation for 5000 replications and do not indicate bias or incorrect convergence. The results are aligned with the adopted nominal levels. The rejection rates are consistently close to the respective levels, and the mean values of the S statistic converge to 1, as expected, given that the statistic's distribution converges to χ^2_1. It can also be observed that the test performs well for small signal lengths and continues to perform well as the data length increases.

The receiver operating characteristic (ROC) curve shown in Fig. 3 illustrates the performance of the proposed change detection model. To generate the ROC curve, a new Monte Carlo simulation with 5000 replications was conducted; in this case, it generated sample sizes $N = N_1 + N_2 = 200$, where $N_1 = 100$ is a signal from $Y_1 \sim \mathcal{R}(\hat{\sigma}_0 = 0.173)$ and $N_2 = 100$ is a second signal generated from $Y_2 \sim \mathcal{R}(\hat{\sigma}_1 = 0.173 + 0.03)$. The Rayleigh parameter was set to $\hat{\sigma}_1 = 0.203$ based on a realistic but moderate shift in the distribution, allowing us to assess how well the method detects subtle changes. This choice was guided by prior studies in SAR images. For each iteration, the PD and the probability of false alarm were calculated and stored, with thresholds varying from $(0, 1)$. Finally, the curve was generated using the average results from all iterations. The area under the curve (AUC) obtained was 0.9019. This value suggests that the proposed detector efficiently distinguishes changes when the data do not follow the same distribution, i.e., different parameters.

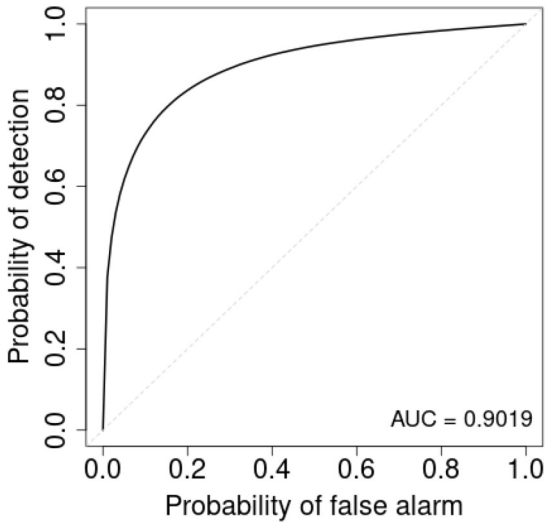
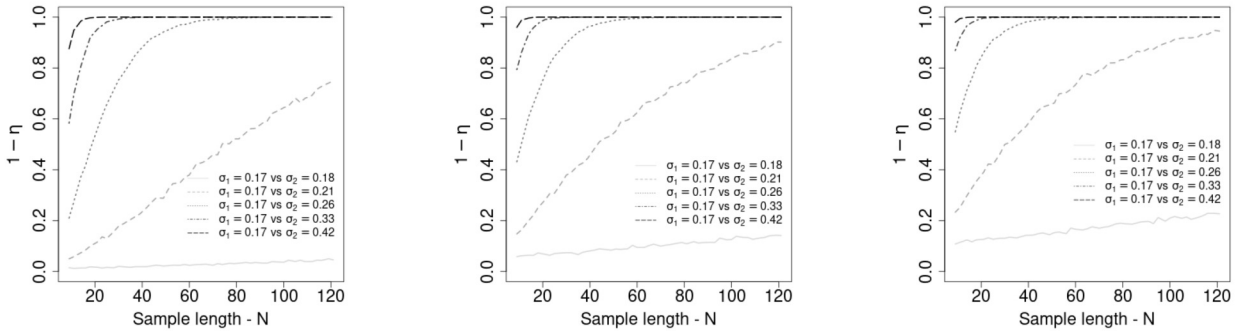


Fig. 3. ROC curve obtained through Monte Carlo simulations using samples from two Rayleigh distributions with scale parameters $\hat{\sigma}_0 = 0.173$ and $\hat{\sigma}_1 = 0.173 + 0.03$ and $n = 200$. The AUC (AUC = 0.9019) indicates strong discriminative performance of the proposed change detector.



(a) (b) (c)

Fig. 4. Simulation of PD for change detection, varying sample sizes and nominal levels. The PD is calculated as $1-\eta$, where η represents the Type II error (false negative). Sample sizes range from $N = 9$ to $N = 121$. For larger shifts in the Rayleigh scale parameter (e.g., $\Delta\sigma = 0.09$ or 0.25), the PD exceeds 0.8 with smaller sample sizes. For moderate changes ($\Delta\sigma = 0.04$), a sample size above 100 is typically needed to reach this threshold—except under the strictest nominal level ($\alpha = 0.01$), where the maximum PD remains closer to 0.7. These results help estimate minimum sample requirements under different detection scenarios. (a) PD for the 1% level. (b) PD for the 5% level. (c) PD for the 10% level.

The PD reflects the effectiveness of the change detector; therefore, a simulation study was conducted to evaluate and analyze this performance aspect. The PD is the probability of rejecting the null hypothesis ($\mathcal{H}_0$) under the alternative hypothesis ($\mathcal{H}_1$) when $\mathcal{H}_1$ is true. The hypotheses are defined in (8). In change detection, the PD can be quantified as the rate $1-\eta$, where η is the Type II error or false negative (FN), defined by $\eta = (T - C^*)/T$, where C^* is the number of rejections of $\mathcal{H}_0$ under $\mathcal{H}_1$ and $T = 5000$ (the number of replications in the simulation).

In this case, it was checked if two samples of $Y_1 \sim \mathcal{R}(\sigma_1 = 0.173)$ and $Y_2 \sim \mathcal{R}(\sigma_2 = 0.173 + k)$, for $k = \{0.01, 0.04, 0.09, 0.16, 0.25\}$, were correctly identified as a situation where a change had occurred. Curves were generated by varying the nominal level, sample sizes, and sample distributions. It was considered 49 sample sizes, varying from $N \in \{9, \ldots, 121\}$. In the context of the image, these sizes can be interpreted as pixel matrices, where 121 observations correspond to an 11×11 window. Fig. 4 contains the plots for the PD curves, grouped by nominal level, where Fig. 4(a)–(c) shown the PD curves for the nominal level of $\alpha = 0.01 = 1\%$, $\alpha = 0.05 = 5\%$, and $\alpha = 0.1 = 10\%$, respectively.

When analyzing the curves for the three nominal levels, it is possible to observe that PD above 0.8 is achieved for small signal lengths, when $\sigma_2 = 0.173 + k$, with $k > 0.04$. When $k = 0.04$, i.e., $\sigma_2 = 0.213 \approx 0.21$, for nominal levels $\alpha = 0.05$ and $\alpha = 0.1$, the PD exceeds 0.8 for sample sizes above 100; otherwise, at the nominal level $\alpha = 0.01$, the maximum PD in the graphs is close to 0.7.

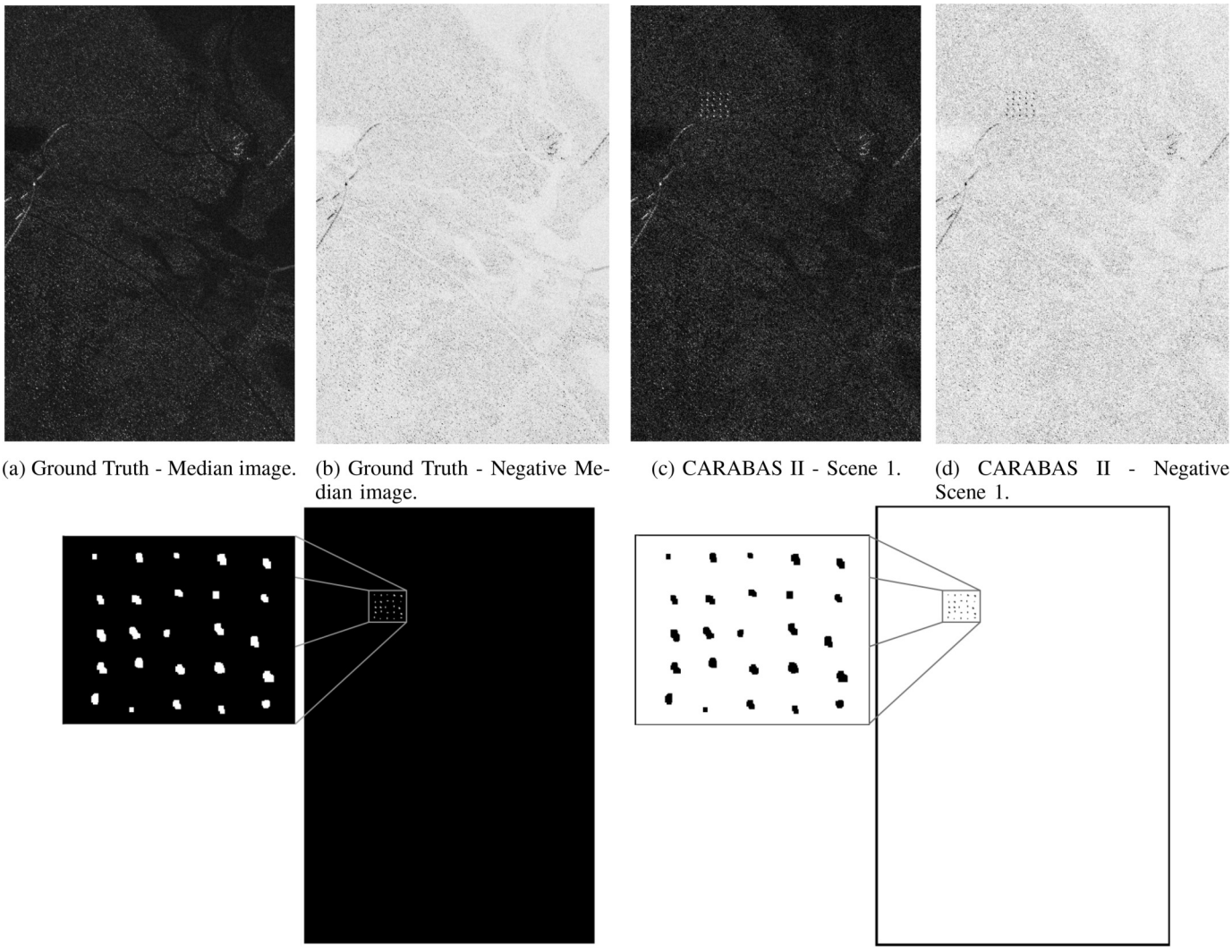
These simulation scenarios serve complementary validation roles. The Type I error analysis checks whether the null hypothesis is correctly retained when true. The ROC analysis assesses the detector's ability to distinguish between distributions when a change occurs under fixed conditions ($\sigma_0 = 0.173$ and $\sigma_1 = 0.203$). Finally, the PD analysis explores this discriminative ability across varying signal strengths and sample sizes. Together, they provide a comprehensive evaluation of the proposed method's robustness and detection capacity. Based on the analyses conducted with the simulations, the change detector demonstrated effective performance, evidenced by its ability to accurately and consistently detect changes in simulated scenarios. The results suggest that the method is robust for detecting variations in image data with similar characteristics to those tested.

B. SAR Image Modeling

In this section, experiments with two measured SAR datasets are presented to demonstrate the applicability of the proposed approach in the SAR image analysis. We employed the introduced method to detect changes in SAR image scenes. The first experiment using real data employs single-look images from the CARABAS II dataset, which has been extensively studied for detecting military vehicles hidden by forests. To further demonstrate the robustness of the proposed model, we also include a multilook image sourced from the UAVSAR radar. This radar was previously applied in a change detection method for single-look polarimetric data, utilizing geodesic distance techniques [53].

1) CARABAS II Data: This experiment was carried out with single-look SAR images captured during a flight campaign in 2002 in northern Sweden, using the CARABAS II system, a Swedish synthetic aperture radar system in the VHF band. This is a VHF wavelength-resolution system, meaning the images exhibit minimal speckle noise. The system operates with HH polarization, and its spatial resolution is 2.5 m in both azimuth and slant range. The CARABASII images are: 1) represented as 3000×2000 pixel matrices (with each pixel measuring 1×1 m), corresponding to a 6 km^2 area and covering a scene of 3×2 km; 2) georeferenced to the Swedish reference system RR92; and 3) available in [66]. More information about the data can be found in [50] and [51].

The CARABAS II database consists of 24 scenes, distinct in three different capture structures, with eight scenes captured under the same flight and illumination conditions [50]. In this article, a random capture structure was selected, and change



(a) Ground Truth - Median image. (b) Ground Truth - Negative Median image. (c) CARABAS II - Scene 1. (d) CARABAS II - Negative Scene 1.

(e) $S_S(\mathcal{R})$ - CARABAS II - Scene 1. (f) $S_S(\mathcal{R})$ - CARABAS II - Negative Scene 1.

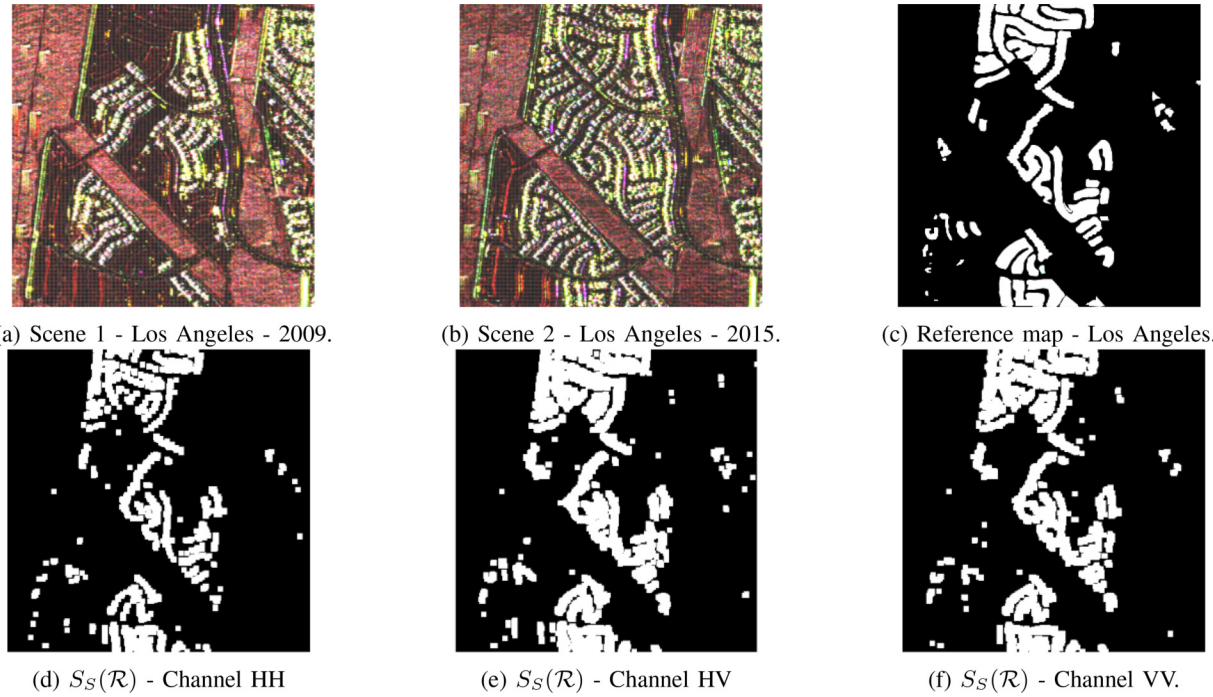
Fig. 5. Change detection results for the CARABAS II SAR data. (Top row) (a) Target-free reference image (median composite), (b) its negative version, (c) one of the target-containing scenes, and (d) its negative version. (Bottom row) Detection maps produced using the proposed method applied to (a) and (c), with (e) standard result and (f) its negative version. Detected changes (targets) are shown in white and black for original and negative images, respectively.

detection was performed in one scene from the eight available in the selected structure. Describing each scene, they contain 25 vehicles, also called targets, in different locations. So, the goal is to apply the change detection method to identify the pixels corresponding to the targets using an image of the scene and a target-free image.

As previously described, the scenes obtained by the CARABAS II system consist of three different structures, each containing eight scenes. Therefore, the target-free image was created from the eight scenes of the selected structure by processing each pixel; the median value for each pixel was obtained, and the final image, called the ground truth, preserves the characteristics of the original scenes while being a target-free scene. This idea was fully explored in [49], where the use of an image stack was employed to compute a ground scene prediction image aiming at removing clutter and noise in the surveillance image. Different methods were tested, and the median-based ground scene prediction was selected as the more reliable free-target image. The ground truth image can be observed in Fig. 5(a). The CARABAS II areas do not have a reference map (RM) and are intended for target detection.

The results obtained for the CARABAS II scenes will be analyzed visually and can be found in Fig. 5. Fig. 5(a) and (c) represents the images used as inputs for the change detector. For each pixel, a 3×3 window centered at the same position in both images is extracted, serving as the input samples. Following as described in Algorithm 1, with the decisions made by the change detector, was made in an image that can be found in Fig. 5(e), representing regions of change (white pixels), and representing areas with no differences (black pixels) between the images in Fig. 5(a) and (c). Fig. 5(b), (d), and (f) presents the negative versions of the images, aiming to provide better visualization of the information and results obtained for the CARABAS II.

The proposed method detected all 25 targets and produced no false alarms. This result is consistent with those reported



(a) Scene 1 - Los Angeles - 2009. (b) Scene 2 - Los Angeles - 2015. (c) Reference map - Los Angeles.

(d) $S_S(\mathcal{R})$ - Channel HH (e) $S_S(\mathcal{R})$ - Channel HV (f) $S_S(\mathcal{R})$ - Channel VV.

Fig. 6. Change detection results on a multilook UAVSAR dataset of Los Angeles captured in 2009 and 2015. (Top row) (a) Image from 2009, (b) image from 2015, and (c) RM of known changes. (Bottom row) Detection results using the proposed entropy-based method for each polarization channel: (d) HH, (e) HV, and (f) VV.

in [50] and [52], where change detection methods were specifically designed to identify these targets. The proposed method also excelled another Rayleigh-based model proposed in [67], with one more detection and two fewer false alarms.

2) UAVSAR Data: The images used for the second experiment are multilook PolSAR data acquired by the UAVSAR sensor developed by NASA's Jet Propulsion Laboratory (JPL) [69]. This radar operates in the L-band at a frequency of 1.26 GHz, with a bandwidth of 80 MHz, providing a spatial resolution of approximately 6×6 m on the ground, equivalent to a pixel covering an area of 36 m^2 [70]. The images were processed with a number of looks (NL) of four, ensuring an improved signal-to-noise ratio for interferometric analyses. For this study, the data were captured during two distinct time periods: April 23, 2009 and May 3, 2015. Fig. 6(a) and (b) represents two images of the same urban area in Los Angeles, USA, in different instances of time.

Fig. 6(c) is an RM illustrating the areas where changes occurred in the Los Angeles region between the two capture periods. For the analysis of this PolSAR image of Los Angeles, the change detector $S_S(\mathcal{R})$ was applied individually to each channel (HH, HV, and VV), considering Fig. 6(a) and (b). Similar to the first experiment, for each pixel, a 3×3 window centered at the same position in both images was extracted, serving as the input samples for the change detector.

Based on the decisions of the detector, images were generated where pixels identified as different are shown in white, while areas considered unchanged are displayed in black. The qualitative results, in terms of visual analysis of the proposed method applied to the Los Angeles region, are presented in Fig. 6(d)–(f).

To confirm the qualitative discussion of predictive maps, six metrics are used to evaluate the performance of the detectors.

1) *False Positive (FP):* Number of pixels indicated as a change by RM, but classified as no change.
2) *False Negative (FN):* Number of pixels indicated no change by RM, but classified as a change.
3) *False Alarm Rate (FA):* (FP+FN)/N, where N is the number of unchanged pixels according to the detector.
4) *Detection Rate (DR):* TP/CG, where TP is the number of pixels indicated as changed by both the RM and the detector, and CG is the number of changed pixels according to the detector.
5) *Kappa Coefficient:* $\kappa = (A - B)/(1 - B)$, where $A = 1 - p_{\text{FP}} - p_{\text{FN}}$ and $B = (p_{\text{TP}} + p_{\text{FP}})(p_{\text{TP}} + p_{\text{FN}}) + (p_{\text{TN}} + p_{\text{FP}})(p_{\text{TN}} + p_{\text{FN}})$, where p_C is the proportion of pixels under the condition C = [TP, FP, FN, TN] relative to the total number of pixels and TN is the number of pixels indicated as no change by both RM and the detector.
6) *MCC Coefficient:* MCC = NUM/DEN, being $\text{NUM}=(\text{TP}\cdot\text{TN})-(\text{FP}\cdot\text{FN})$ and $\text{DEN}=((\text{TP}+\text{FP})\cdot(\text{TP}+\text{FN})\cdot(\text{TN}+\text{FP})\cdot(\text{TN}+\text{FN}))^{1/2}$.

Regarding the performance metrics, the results are presented in Table II. Results were generated for each polarization channel (HH, HV, and VV). This allows for evaluating which channel in this study area is most suitable for achieving better change detection.

Nascimento et al. [32] and Bouhlel et al. [68] also proposed change detectors and applied them to images of the Los Angeles region, using methods based on information theory for the development of the methods. The methods proposed in both studies utilized all polarization channels (HH, HV, VV,

TABLE II

PERFORMANCE EVALUATION OF CHANGE DETECTION METHODS FOR THE LOS ANGELES UAVSAR DATASET. THE PROPOSED METHOD ($S_S(\mathcal{R})$) WAS TESTED ON INDIVIDUAL POLARIZATION CHANNELS (HH, HV, AND VV) USING AMPLITUDE DATA, WHILE PREVIOUS METHODS USED ALL CHANNELS AND INTENSITY DATA. BOLD CELLS HIGHLIGHT THE BEST PERFORMANCE PER METRIC AMONG THE PROPOSED CONFIGURATIONS

Detectors	Channel	FP (%)	FN (%)	FA (%)	DR (%)	κ (%)	MCC
S_{LR} [32]	all channels	0.06	13.43	13.49	20.41	24.56	-
S_{KL} [32]	all channels	0.05	14.06	14.11	19.77	23.02	-
S_S [32]	all channels	0.34	5.47	5.82	35.39	52.48	-
S_R [32]	all channels	0.43	3.71	4.14	42.99	62.27	-
DRT [68]	all channels	-	-	-	63.38	-	-
HLT [68]	all channels	-	-	-	52.83	-	-
$S_S(\mathcal{R})$	HH	4.13	2.82	8.08	79.86	72.22	0.72
	HV	1.72	5.82	9.35	69.99	73.79	0.75
	VV	2.12	4.60	8.17	74.13	75.69	0.76

HH-HV, HH-VV, and HV-VV) as input information for the change detector and considered the data in the intensity structure ($I = A^2$). The visual representations of the compared methods' results are not included in this manuscript but can be found in their respective cited publications. Our proposed change detector uses a single polarization channel and amplitude (A) data, demonstrating competitive performance.

Four change detectors based on hypothesis testing were used by Nascimento et al. [32] with the scaled complex Wishart distribution. They utilized the following statistics: 1) likelihood ratio (S_{LR}); 2) KL distance (S_{KL}); and 3) statistics based on Shannon (S_S) and Rényi (S_R^{β}) entropies.

Bouhlel et al. [68] proposed a detector using the determinant ratio test (DRT) statistic, also based on the scaled complex Wishart distribution, deriving its distribution as Wilks's lambda of the second kind, expressed through Meijer G-functions. The method was compared with the Hotelling–Lawley trace (HLT) and the likelihood ratio test (S_{LR}).

To compare the results, it is essential to have at least one common metric; in this case, the DR metric is available and will be used to compare the methods proposed in this study with those from other studies. Based on the observed results, the best DR value from the cited studies was 63.38%, achieved using the DRT method proposed by Bouhlel et al. [68]. When considering the proposals of this article, $S_S(\mathcal{R})$, the lowest DR value was 69.99%, obtained with the HV channel, followed by the VV channel with a DR of 74.13%, and the highest DR value was achieved using the HH channel, with a DR of 79.86%.

Based on the analysis of the kappa coefficient (κ), the channel with the best performance was VV, with a κ of 75.69%, followed by the HV channel and then HH, with κ values of 73.79% and 72.27%, respectively. The MCC metric also aligned with the kappa coefficient, indicating that the VV channel, with a value of 0.76, achieved the highest accuracy in correctly detecting change points, followed by the HV and HH channels, with values of 0.75 and 0.72, respectively.

VI. CONCLUSION

This article proposed a change detector based on Shannon entropy with the Rayleigh distribution. The proposed method effectively detected changes between two samples through hypothesis testing. Through simulated experiments using information derived from real SAR data, the method achieved favorable results in analyzing Type I error rates consistent with nominal levels, even for small sample sizes.

In addition, the ROC curve and the AUC in simulated data (AUC = 0.9019) demonstrated the high efficiency of the change detector, validating the model's ability to identify changes when the data exhibit distinct distributions. Similarly, the PD analysis indicated that the method effectively rejects the null hypothesis when the alternative hypothesis is true, even under varying sample size configurations.

Considering the experiments with real data, the results demonstrated that the change detector based on the Rayleigh distribution is a viable method for this type of data. The use of different configurations for windowing and nominal levels ensures that the methodology can be adapted to a wide range of applications, reinforcing its contribution to the context of change detection in SAR data.

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